Proposal is **Experimental** ☐ **Computational/Theoretical** ☑ **Hybrid** ☐

Objective

We aim to determine if industry-generated foundation models should be pursued for use in scientific applications that are relatively data-poor and lack generalizability. If successful, our approach could provide near-immediate improvements for a multitude of applications, with minimal investments from LANL and domain experts.

Problem and Proposed Innovation

Companies have spent hundreds of millions of dollars developing "foundation models" (FMs) that have resulted in groundbreaking technologies like ChatGPT (conversational), AlphaFold (protein structure discovery), Codex (Github's "copilot"), and Midjourney (image generation). These FMs feature O(1-10 billion) parameters and vast training datasets made possible only through massive, focused investments. Because FMs have such rich architectures and massive training sets, they can generalize in unprecedented ways, as evidenced by the aforementioned techs. However, the use of FMs for *scientific prediction* remains an unexplored, wild west. Here, we want to understand the degree to which we can exploit multimillion dollar FMs toward improving scientific predictive skill. **How much value do FMs offer for scientific prediction?**

In scientific applications, "foundation model" is often defined differently as a "large-ish model trained on a large-ish dataset" representing a single process or problem. However, rarely–if ever–do scientific problems have nearly the quantity of data available required to train true FMs, leading to a lack of generalizability and suboptimal performance. FMs have learned from a variety of data types (e.g., finance, weather, health care, etc.) that have the potential to restore generalization to data-limited scientific applications. Scientific AI models also tend to suffer from an inability to transfer to another domain, leading to "one model per problem" scenarios. **Within the framework we will propose, FMs offer the potential to overcome both generalizability and transferabilty problems currently hampering progress in AI for scientific advancements.**

FMs have been developed for different data modalities—namely sequential, image, and text–allowing us to envision crafting a general framework that can bring generalizability to a wide range of data-limited, scientific problems. Our idea is to build FUSE: *Foundation model Utilization in Stacked Ensemble learning*. an ensemble learning framework where FMs constitute a "level-0 learner" bloc (Figure 1), followed by a "level-1 learner" that synthesizes the outputs of the FMs into a final prediction. While the structure of FUSE is relatively simple, it stitches together two novel concepts: FMs as useful interpreters of scientific data (boosting generalizability) and ensemble learning (boosting performance). The framework itself is adaptable, as adding additional FMs may be embedded, or existing ones replaced, preventing "model fossilization" by allowing adaptability to new FMs as they become released. FUSE also offers huge potential for data fusion and augmentation techniques. For example, models built for flooding (i.e. numeric data) currently have no way to learn from news reports (i.e. text data) that may describe flooding extents. **Our approach would open new possibilities toward fully exploiting information from data modalities that have historically not even been considered relevant.**

R&D Methods and Anticipated Results

The Application We will test FUSE on the problem of daily streamflow prediction, which is important for a host of national security issues ranging from flooding to water security to hydroelectricity generation. This application has three main benefits: 1) our team has years of expertise

studying daily streamflow modeling, including a suite of AI models we can benchmark FUSE results against [1, 2], 2) we have assembled the largest synthesis of streamflow training data in existence [3] (over 40,000 locations with daily time series) so that very little time will be spent on data curation, and 3) streamflow modeling is well-suited to using multi-modal predictive features as inputs. Inputs to AI streamflow models are daily meteorologic data (precipitation, temperature, etc.) and characteristics of watersheds (slope, vegetation, soil types, etc.). Traditionally, these input features are averaged across each watershed into a 1-D time series vector. However, FUSE will allow us to use, for example, FMs that can interpret 2-D images of weather fields rather than simple averages—to the best of our knowledge, no published research has taken this approach to the problem, nor have we seen attempts to exploit FMs for streamflow modeling more generally.

Evaluation: FUSE streamflow predictions will be compared against our existing benchmarks. In particular, we will evaluate (1) samples (watersheds) for which the benchmark models struggle—namely dry, "flashy" watersheds—to evaluate performance boosting and (2) FUSE's ability to perform in watersheds to which it has not been exposed to test generalizability. The results from these comparisons will provide a sense of (1) the degree of "universality" FMs offer, (2) potential benefits of ensemble learning, and (3) whether FUSE may be a powerful approach for other applications.

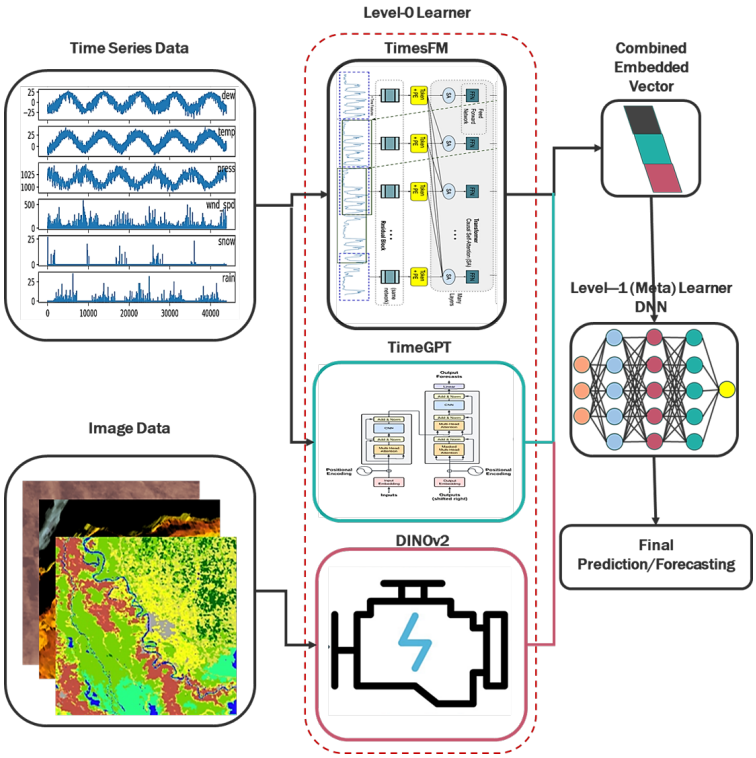

Fig. 1: The proposed FUSE with dual-levels of learning. The level-0 learner consists of three large scale foundation models, namely, TimesFM, DINOv2, and TimeGPT. The level-1 or meta learner consists of a deep neural network (DNN).

The Framework We will develop a dual-level stacked generalization ensemble model (Figure 1), combining three pre-trained foundation models as level-0 learners and a deep neural network (DNN) as the level-1 meta-learner. We will develop a unique training strategy to avert any potential overfitting, where training datasets will be divided into three parts. *Part A* will be used to fine-tune the foundation models in the level-0 learner. *Part B* will draw inferences from the fine-tuned level-0 learners and use the outputs to train the DNN in the level-1 learner. Finally, *Part C* will serve as a test set for evaluating the performance of the combined ensemble learning model.

Level-0 Learner: Our proposed approach leverages three powerful FMs: TimesFM [4], DINOv2 [5], and TimeGPT [6]. TimesFM is a decoder-only type pre-trained transformer model trained on a vast dataset of 100 billion real-world time series tokens from various sources including Google Trends and Wikipedia Pageviews. The model will receive the time series data of precipitation, temperature, wind speed, etc., segregate the series into smaller segments, apply positional encod-

ing, and generate at the output of stacked transformer layers an embedded vector that captures essential intrinsic features of the inputs. This resultant vector is provided to the subsequent DNN model (level-1 learner) for further processing. The DINOv2 is a Vision Transformer (ViT) model pre-trained with 142M images utilizing self-supervised learning. DINOv2 will process the satellite images of the region so that images are segmented into patches, then encoded and processed by the in-built transformer units and generate a feature vector holding latent information relevant to streamflow prediction. The generated feature vector will act as an input for further processing by the level-1 learner, i.e., the DNN model. TimeGPT is a transformer-based time series forecasting model that can predict complicated and multivariate data. After processing, the output from the TimeGPT model would flow into the subsequent DNN of the level-1 learner. Since TimesFM and TimeGPT are FMs that process time series, their combination enhances the ensemble's predictive accuracy by availing diverse, nonlinear temporal patterns to the level-1 learner.

Level-1 Learner: The level-1 learner synthesizes the outputs of FMs into a final prediction. We will initially use a DNN with cross-validation and regularization techniques to capture the nonlinear relationships between the outputs of the FMs while preventing overfitting. Depending on trainability and performance, we may test other machine- and deep-learning architectures as the level-1 learner.

Uncertainty: Scientific applications undoubtedly call for uncertainty quantification (UQ), and we will build UQ into FUSE by modeling outputs as an independent Normal distribution and learn to predict its (feature-dependent) mean and variance (e.g. [7–9]. Variance thus serves as a UQ estimate on each FUSE prediction, and the loss function (typically MSE) becomes a weighted MSE that maximizes the log-likelihood of the output Normal distribution. This approach is general to FUSE modifications and applications. FUSE also provides a framework for quantifying epistemic (i.e. model) uncertainty and assessing FM-based meta-learning with UQ.

The Risk We see the most significant risk of FUSE as finding the right level-1 learner architecture that will effectively exploit the abstractions of the level-0 FMs and without overfitting. If the proposed DNN results in underfitting, we will test other deep learning regression models such as 1-D CNNs and XGBoost. Our level-1 training will be performed on unseen data to limit overfitting. It is also possible that some FMs do not interpret scientific data in useful ways. This will be tested by removing FMs one at a time and comparing performances. This kind of FM "failure" would be highly valuable towards understanding the limitations of FMs for scientific problems.

Importance for the Laboratory and the Nation

If FUSE is successful, its implications could be extremely beneficial to almost all aspects of LANL's mission as well as dozens of scientific domains. Financially, we could exploit millions of dollars in private FM investments to improve scientific prediction for our sponsors with relatively small investments of our own. Our application to streamflow prediction would have immediate impacts with ongoing sponsors within the Climate Impacts to National Security program and DOE Office of Science projects *InteRFACE* and *NGEE Arctic*. More broadly, LANL's latest Agenda has made clear its priority to invest in AI technologies and workforce development, and the ideas in this proposal could amplify the impact of much of the work being done as part of the AI Strategic Institutional Commitment. Our work would posture LANL to propose a much larger effort from the (hopefully) forthcoming DOE Frontiers in AI for Science, Security, and Technology (FASST) initiative.

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Glossary of Acronyms

AI - Artificial Intelligence
DNN - Deep Neural Network
1D CNN - 1-Dimensional Convolutional Neural Network, a type of convolution neural network (CNN) suitable for processing 1-dimensional data
ER - Exploratory Research, an LDRD funding mechanism
FASST - Frontiers in AI for Science, Security, and Technology, a DOE initiative poised to make major AI research investments
FM - Foundation Model, in particular those with billions of parameters and trained on many data types
FUSE - Foundation model Utilization in Stacked Ensemble learning, the proposed innovation
GPU - Graphics Processing Unit, type of processor hardware
HPC - High Performance Computing, supercomputing
InteRFACE - Interdisciplinary Research for Arctic Coastal Environments, a multi-institution DOE project
LDRD - Lab Directed Research and Development
ML - Machine Learning
MSE - Mean Squared Error
NGEE - Next Generation Ecosystem Experiments, a series of ongoing major DOE investments focused on various global regions
SSL- Self-Supervised Learning, a machine learning paradigm where the model learns using the data itself rather than an external human provided output label
TimesFM - Time Series Foundation Model, a pre-trained foundation model for time series analysis
TimeGPT - Time Generative Pre-trained Transformer, a pre-trained foundation model for time-series analysis
UQ - Uncertainty Quantification
ViT - Vision Transformer, a kind of transformer architecture to process images
VotE - Veins of the Earth hydrology data platform
XGBoost- Extreme Gradient Boosting, is a scalable, distributed gradient-boosted decision tree (GBDT) machine learning library

Appendix: Qualifications of PI and Team

In addition to being potentially highly valuable across LANL problems and being aligned with LANL investments, this project belongs at LANL because of the unique expertise our team brings. Our team is intimately familiar with the successess and limitations of applying AI methods to complex, environmental systems relevant to national security and LANL interests [1, 10–12]. Furthermore, our LANL team has developed the most comprehensive streamflow training dataset in existence [3]. We also have demonstrated capability to design new ML/AI architectures [13] and deploy UQ solutions to deep learning models [7–9]. We also anticipate using unique LANL HPC resources, including Darwin, NagiosXI (maintained by EES-16), and potentially Chicoma systems. The three performers and their roles for this proposal are described below.

Jon Schwenk (PI, Scientist 3, A-1)
Jon's research interests have focused on watershed hydrology and streamflow prediction using both physics-based and machine-learning approaches. He has spent most of his LANL career bringing together AI/data science and domain expertise to tackle environmental science problems. Jon has PI'd four projects of >$1 million and has had leadership roles on many more. He will provide the expertise to ensure that AI development is consistent with our physical knowledge of watershed processes, and as the lead developer of the streamflow data platform "VotE," Jon will handle all data needs. He will also guide the development of the project during all phases.

Shubhendu Kumar Singh (Co-PI, Postdoc, EES-14)
Shubhendu obtained his Ph.D. in Automotive Engineering from Clemson University. His core expertise lies in Artificial Intelligence (AI), computer vision, Physics-Informed Machine Learning, and complex dynamic systems. He has worked on various projects at the intersection of AI and Engineering Sciences supported by DARPA, ONR, and CESMII (DOE). His current research at LANL focuses on developing reinforcement learning algorithms, to optimize water reservoir operations.

Shubhendu began his postdoc appointment in August, 2024, turning down other opportunities to work with our team on relevant and important environmental problems. He enthusiastically intends on remaining at LANL for the full tenure of his appointment and represents a very low-risk for departure before this project would be completed. In the event of an unexpected departure, Jon and Cristina would tap into their networks across EES, CCS, T, and A Divisions to find a postdoc with the skillset and availability needed to help develop FUSE. Shubhendu will be in charge of implementing the FUSE framework, fine-tuning the FMs, training the level-1 learner, performing performance testing, and porting all necessary data and code to HPC.

Cristina Garcia Cardona (Co-I, Scientist 3, CCS-3)
During her time at LANL, Cristina has taken part of several DOE ECP, DOE ASCR, LDRD ER, DR and DI projects. Her research focuses on developing deep learning models for inverse problems in imaging, uncertainty quantification for regression problems, and reinforcement learning. She has extensive experience with scientific software implementations, including CANDLE (CANcer Distributed Learning Environment) a DOE ECP RD 100 winner in 2023, and SCICO, a JAX-based Scientific Computational Imaging COde. Cristina will be responsible for the design and testing of FUSE's UQ and will contribute to FUSE's overall design.